Firewall BasicsTheory

1.Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

ANS:

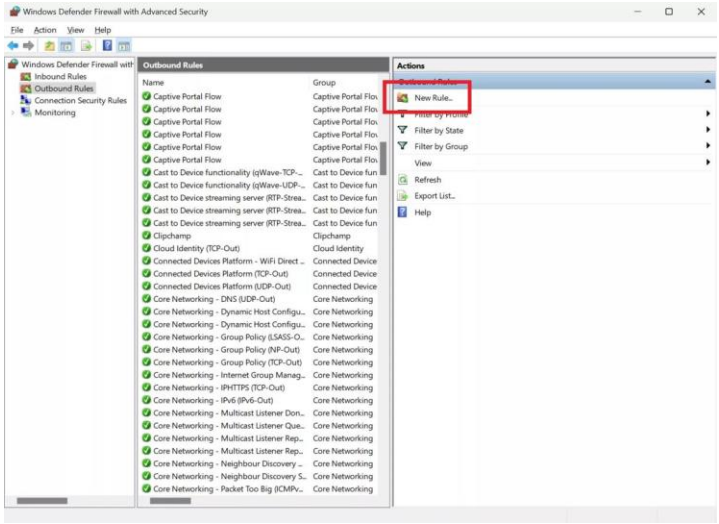
Block Spotify using Windows Defender Firewall

I can guide you through creating an outbound rule in Windows, but I can't directly change your computer's firewall settings from this chat.

Step 1: Open Firewall settings

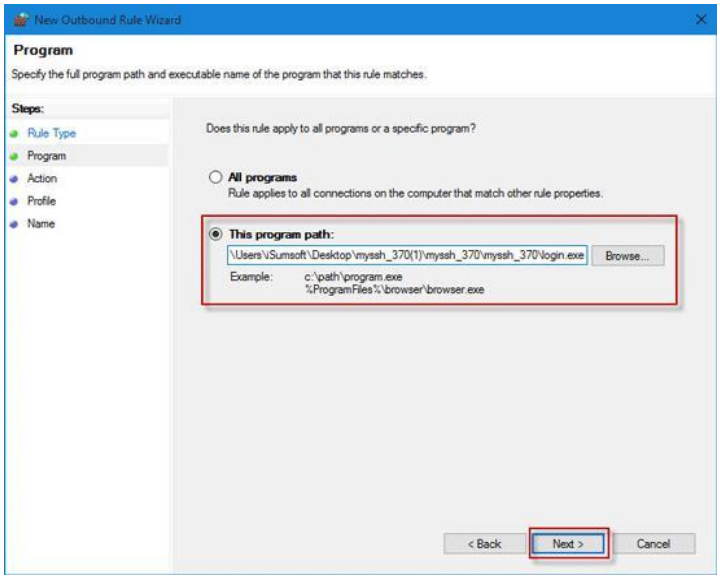
1. Press Win + R.
2. Type wf.msc and press Enter.
3. Select Outbound Rules in the left panel.
4. Click New Rule... in the right panel.

Step 2: Create the Spotify blocking rule



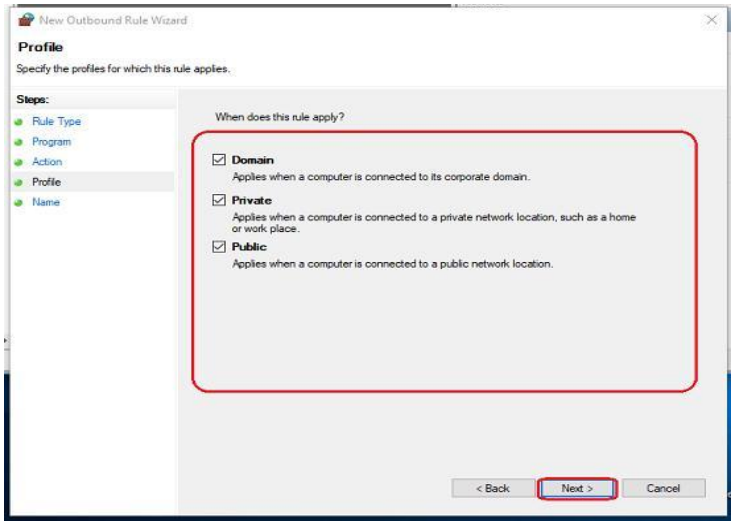
1. Select Rule Type

Choose Program, then click Next.



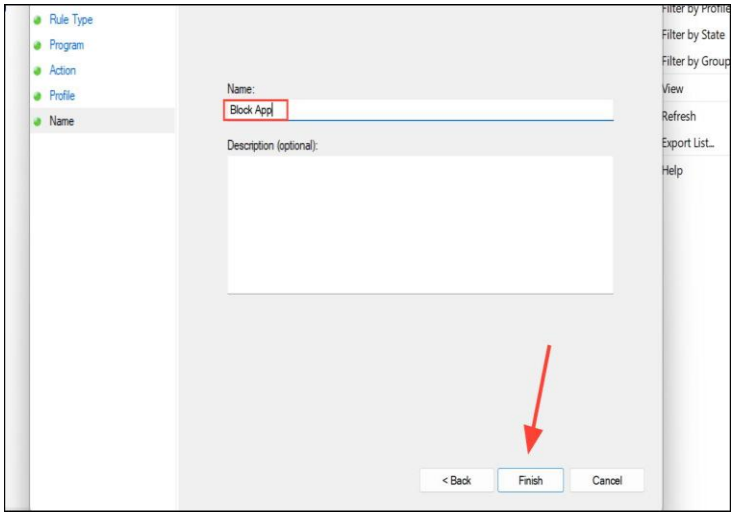
2. Select Spotify

Choose This program path and browse to Spotify's desktop executable, usually Spotify.exe. Its location depends on how Spotify was installed.



3. Block the connection

Select Block the connection, click Next, and select Domain, Private, and Public.



4. Name the rule

Enter Block Spotify Outbound Traffic and click Finish.

Step 3: Verify the rule

- Open Outbound Rules and find Block Spotify Outbound Traffic.
- Confirm the rule is enabled.
- Open Spotify and try playing a song. Online streaming should fail if the rule targets the correct executable.

Result: Windows Firewall should block outbound network connections made by the selected Spotify executable. Other applications should continue to work normally.

2.Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

ANS:

Create an Inbound Rule for Google Chrome on Port 443

Follow these steps in Windows Defender Firewall with Advanced Security.

Step 1: Open Firewall settings

1. Press Win + R.
2. Type wf.msc and press Enter.
3. Select Inbound Rules on the left.
4. Click New Rule... on the right.

Step 2: Configure the rule

1. Select Rule Type

Choose Custom and click Next.

2. Select Google Chrome

Under Program, choose This program path and browse to chrome.exe. A common location is:

C:\Program Files\Google\Chrome\Application\chrome.exe

3. Set the port

Under Protocol and Ports, select TCP and set Local port to 443. Leave Remote port as All ports.

4. Allow the connection

Select Allow the connection. Choose the required profiles: Domain, Private, and/or Public.

5. Name the rule

Enter Allow Chrome Inbound TCP 443 and click Finish.

Step 3: Verify the rule

Open Inbound Rules, locate Allow Chrome Inbound TCP 443, and confirm that it is enabled.

**3.Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.**

ANS:

Steps to Block a Specific IP Address Using Windows Defender Firewall

Example IP address: 123.45.67.89

1. Open Windows Defender Firewall: Press Win + R, type wf.msc, and press Enter.

2. Create a new rule: Select Outbound Rules from the left panel, then click New Rule on the right.

3. Choose the rule type: Select Custom and click Next.

4. Select the program: Choose All programs to block traffic from all applications, then click Next.

5. Set protocol and ports: Keep the default settings to apply the rule to all protocols and ports, then click Next.

6. Configure the Scope tab: Under Which remote IP addresses does this rule apply to?, select These IP addresses and click Add. Enter 123.45.67.89, then click OK and Next.

7. Block the connection: Select Block the connection and click Next.

8. Select profiles: Check Domain, Private, and Public, then click Next.

9. Name the rule: Enter Block IP 123.45.67.89 and click Finish.

10. Verify the rule: Confirm that the new rule appears under Outbound Rules and is enabled.

4.Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

ANS:

Hardware Firewall vs Software Firewall

Feature	**Hardware Firewall**	**Software Firewall**
1. Location	Separate physical device placed between the network and internet.	Installed on an individual computer or server.
2. Protection	Protects multiple devices across a network.	Protects a specific computer or device.
3. Cost and management	Usually more expensive and requires network configuration.	Usually cheaper and easier to configure.

Use Cases

- Hardware Firewall: A large office uses a hardware firewall to protect its entire company network from unauthorized access and cyberattacks.
- Software Firewall: A home user uses Windows Defender Firewall to block suspicious connections and control applications on their PC.